One Record per Click: Offline-Teacher Soft Targets for Delayed Feedback under a Single-Visit Constraint

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Abstract. Conversion-rate models must be trained before their labels exist. Most published corrections for this delay assume the training stream can be revisited — a record is written at click time and repaired later by duplicating the event, relabelling it, or adding heads at further horizons. An append-only pipeline cannot do that cheaply. We impose a *single-visit* constraint — at most one training record per click, its label frozen when written — and measure how many published methods satisfy it: reproducing 11 of them from scratch on the Criteo conversion logs at a one-day attribution window, 8 write more than one record for at least one click, up to 4.00 on average and 97 at worst. What survives is a narrow design space in which the only remaining choice is the target the single record carries. We study four points on that target-hardness axis across five served capacities (1,890–9,580,477 parameters) and three pipeline write delays. Waiting for the matured label beats training on the censored one by a wide margin. Replacing part of the hard label with an offline teacher's estimate improves on waiting in all 15 cells; removing the hard label entirely does not, crossing from better than waiting to worse as capacity grows. The hybrid target matches an online label corrector to within noise rather than beating it, so the case for an upstream teacher is architectural: it is what lets K objectives with K different attribution windows share one record written at one time. Freezing the teacher into a build-time artifact costs +0.00092 NE, and nothing at all at write delays of 30 minutes or more.

Keywords: Conversion rate prediction · Delayed feedback · Knowledge distillation · Online learning.

1 Introduction

A conversion-rate model must be trained on events that have not happened yet. The click is observable at once; the conversion it may cause arrives minutes, hours or days later. Wait for the attribution window to close and the model is always stale by its width. Do not wait, and every conversion still in flight is recorded as a negative.

A long line of work corrects this, from an explicit model of the delay distribution [1] through importance weighting [15], elapsed-time sampling [14], reingesting an event once its outcome is known [6,5], unbiased treatment of the fake-negative population [2], multiple prediction horizons [7,9,16], and corrective updates from influence functions [4]. The statistics differ; but realised as streaming pipelines, many of them need the training stream to be something one can go back and add to.

We are interested in the case where it is not. In an append-only log a record is written once, at a fixed age after the click, carrying whatever the label is then, and never rewritten. Revisiting such a click means either mutating its label — which turns a stream into a database and makes a reader's view depend on when it read — or appending a second record. Appending is cheaper to build but not free statistically: a click whose conversion arrives late then appears more often than one that never converts, so the fitted example distribution no longer matches the traffic, and the multiplier hits training cost at every step. Call the property that avoids both *single-visit* — at most one training record per click, its label frozen when written. It is a restriction one may reasonably pay to escape; we ask only that the price be visible.

We reproduce 11 delayed-feedback methods from the literature from scratch on the public Criteo conversion logs and count the training records each one's served model consumes per click, at a one-day attribution window. 8 write more than one — up to 4.00 records per click on average, and 97 for a single click at worst. Only FSIW, ULC, TWICE are single-visit.

What survives is narrow enough to be informative: once a method may write one record, the only freedom left is *what label it carries* — a hard label that is wrong (the censored outcome), one that is right but late (wait for the window), or a target partly or wholly supplied by a model. We instantiate four points on this *target-hardness* axis and ask what each degree of softness buys at fixed single-visit cost.

Waiting beats the censored label by a wide margin at short write delays, so the status quo is no straw man. Replacing part of the hard label with an offline teacher's estimate lowers NE against waiting in all 15 cells (10 significantly); removing the hard label *entirely* does not, helping only at the smallest capacities and losing to waiting at the largest. And the hybrid matches an online label corrector [12] to within noise — which matters less for accuracy than for where each puts the machinery. A model serving several conversion objectives must reconcile attribution windows differing by orders of magnitude. An online corrector entangles each window with the served model's own training loop; a teacher moves it upstream, leaving the served pipeline to consume records that carry a number and no window at all. We also measure how much of the teacher must stay live: freezing it after initialisation, so that it becomes a build-time artifact rather than a running service, costs +0.00092 NE on average. This is a precondition for consolidating many objectives, not a demonstration of it; Sect. 6 states what the evidence supports.

2 Setting and the Single-Visit Constraint

2.1 Streaming protocol

A click i occurs at time C_i with features x_i and, if it converts, does so after a delay $D_i > 0$ ($D_i = \infty$ otherwise). An *attribution window* v fixes the business target, $Y_i^{(v)} = \mathbf{1}[D_i \le v]$, knowable only at $C_i + v$. A streaming pipeline cannot wait that long, so it writes the record at a *write delay* $o \ll v$ carrying the outcome visible then:

$$Y_i^{(o)} \;=\; \mathbf{1}[D_i \le o] \;\le\; Y_i^{(v)}. \tag{1}$$

$Y^{(o)}$ is never a false positive and often a false negative; how often is $F(o) = P(D \le o \mid D \le v)$, the share of the target's positives visible at write time. At $v = 1\,\mathrm{d}$ on our corpus $F(o) = 0.355$, 0.688, 0.803 for $o = 5$, 30, 90 minutes: at a five-minute pipeline two thirds of the conversions that will count are still missing. Training runs in hourly rounds — consume the records released since the last cut-off, then score every click of the next hour once, retrospectively against $Y^{(v)}$ — so every method is judged on the true business label whatever it trained on, and update-then-predict keeps the evaluation causal.

2.2 Two admissibility rules

Rule 1 (causality). A record's label must be computable when the record is written and must not change afterwards. Labelling a record $Y^{(v)}$ while writing it at $C_i + o$ uses an outcome that has not happened — an upper reference (ORACLE), not a system. Strict.

Rule 2 (single-visit). Writing n_i for the number of times click i enters the served model's loss over its lifetime, Rule 2 requires $\max_i n_i \le 1$: a strong practical constraint, being the property that decides whether a method drops into an append-only pipeline unchanged.

The criterion is $\max_i n_i$, not the mean: it is the *existence* of a second write, not its frequency, that obliges the pipeline to support one. And the rule governs the *served* model's stream: auxiliary estimators (importance weights, delay heads, teachers) are not charged, because the served stream is what the pipeline must materialise. Under a stricter reading, in which any auxiliary fitted on matured labels counts as a second visit, the screen would eliminate FSIW, ULC *and both of our own soft-target arms* alike.

2.3 The screen

Table 1 reports, for 11 methods reproduced from scratch under one protocol, the training records per click their served model consumes at $v = 1\,\mathrm{d}$.[1]

[1] Two measurement traps, both of which understate duplication. Dividing records consumed by records released conflates duplication with traffic variation, since a maturity-gated stream reads clicks from v ago when the click rate differed; the denominator must be *distinct clicks touched*. And a window shorter than a method's

Table 1. The single-visit screen at $v = 1\,\mathrm{d}$. *Records/click* is the lifetime number of training records the served model consumes per distinct click; *max* is the largest for any single click. 8 of 11 methods write more than one.

Method		Train rows	Records/click	Single-visit
DFM	[1]	2,451,445	1.33 (max 8)	×
FSIW	[15]	4,341,691	1.00	✓
ES-DFM	[14]	4,482,378	1.03 (max 2)	×
ULC	[12]	4,341,691	1.00	✓
DEFUSE	[2]	4,482,378	1.03 (max 2)	×
DDFM	[3]	8,217,290	1.89 (max 2)	×
IF-DFM	[4]	12,092,149	2.79 (max 97)	×
FTP	[7]	17,366,764	4.00 (max 4)	×
MISS	[9]	13,236,870	3.05 (max 5)	×
PI	[16]	8,683,382	2.00 (max 2)	×
TWICE	[8]	4,341,691	1.00	✓

8 of 11 methods write more than one record for at least one click, and the factors are not marginal: FTP consumes 4.00 records per click, MISS 3.05, and IF-DFM reaches 97 writes for a single click, its mechanism being an update applied whenever a label reverses. Only FSIW, ULC, TWICE are single-visit. The plain re-ingestion baseline — write the click as a negative, write a positive when the conversion lands [6,5] — is not a named correction and so is not listed, but our implementation measures 1.03 records per click and fails the same criterion.

Of the survivors, FSIW corrects the *loss* and TWICE adds a head on the conversion clock; both leave the record's target untouched. ULC alone changes the target, making it the comparator for the axis we study next — not because the other two are inferior, but because varying three axes at once would attribute nothing.

3 A Target-Hardness Axis

Under Rule 2 a method gets exactly one record per click, so the design space collapses to one question: what target $\tilde{y}_i$ does that record carry? Table 2 lists the four points we study; the pipeline, schedule, features and optimiser are identical across them.

FRESH is the uncorrected floor: by Eq. (1) it trains on a label wrong for a $1 - F(o)$ share of eventual positives. WAIT defers the write until the outcome is real, so its label is correct but the freshest example it has ever seen is v old; a practitioner who refuses both to duplicate records and to train on wrong labels

revisit horizon cannot see the revisit, so we score only clicks whose entire horizon lies inside the stream. Duplication is a property of emission logic, not of the model: re-running the audit at the smallest and largest capacities of Sect. 4 gives identical counts.

Table 2. The four points of the target-hardness axis. Δ is the age at which the record is written. Only WAIT defers the write; the rest emit at o and differ in what they put in the label field.

Arm	Target $\tilde{y}_i$	Δ	Hard label
FRESH	$Y_i^{(o)}$	o	censored
WAIT	$Y_i^{(v)}$	v	true, late
CORRECT-DISTILL	$Y_i^{(o)} + (1 - Y_i^{(o)})\, w^T(x_i)$	o	positives only
DISTILL-ONLY	$\hat{p}_v(x_i)$	o	none

is already running it. WAIT is also the same kind of object as the teacher below — a model fitted on matured records — so this paper's central contrast reads as one question: is it better to *serve* the matured model, or to distil it into a student riding the fresh pipeline?

CORRECT-DISTILL keeps the realised label and softens only the observed negatives,

$$\tilde{y}_i \;=\; Y_i^{(o)} + (1 - Y_i^{(o)})\, w^T(x_i), \qquad w^T(x) \approx P\Big(Y^{(v)}{=}1 \,\Big|\, Y^{(o)}{=}0,\, x\Big)\,, \tag{2}$$

so an observed positive is certain and stays exactly 1. This is the functional form of ULC [12]; what differs is the estimator, and because the form is shared exactly, a comparison between them isolates the estimator and nothing else. DISTILL-ONLY goes further and regresses on $\hat{p}_v(x) \approx P(Y^{(v)}{=}1 \mid x)$ alone: the record then carries features and a number, and no observation of what happened.

3.1 The teacher

The teacher is a separate model trained on the *matured* stream — a record enters it at $C_i + v$ carrying $Y_i^{(v)}$, so its own labels are correct and frozen, and it measures 1.00 records per click under the audit of Table 1. It carries $\hat{p}_v$ and w^T as two read-outs on one trunk, and the student consumes one of them. Crucially it is *evaluated* at the fresh release: at $C_i + o$ the student queries it for x_i and freezes the resulting number into the record. The target is therefore knowable when the record is written and never revised, so Rule 1 holds even though the target is soft: softness is not an exemption from causality, only a different way of satisfying it. The construction is an instance of distillation from privileged information [13]: the teacher sees the matured label, which is precisely what the student cannot see at write time, and transfers it as a number rather than as an observation.

Teacher capacity is fixed at 9,456,897 parameters for every student. This is deliberate: the teacher is training-only, so its size costs nothing at serving time, but it means *served* capacity is matched across arms while total training capacity is not. At our largest rung student and teacher are comparable; at the smallest the teacher is three orders of magnitude larger.

The teacher is *decoupled* — its parameters are not in the served model's optimiser — but it need not be *frozen*: our default teacher keeps learning from the matured stream, so it is a live service, whereas one frozen after causal initialisation would be a build-time artifact — cheaper to operate, less informed. We run both (Sect. 5).

ORACLE, our upper reference, writes at $\Delta = o$ but labels the record $Y_i^{(v)}$: it violates Rule 1 by construction and bounds what a correction could recover — in our setting in all 15 cells but one, where the difference is not significant, though this is not guaranteed in general.

4 Experimental Setup

Data and protocol. We use the public Criteo conversion logs released with [1], the standard corpus for this problem. Clicks span 60 days. Days $[0, 30)$ initialise the model causally; days $[30, 60)$ are the streaming period, divided into 719 hourly rounds. At each round the model trains on the records released in the preceding hour and then scores every click of the following hour exactly once, giving 8,196,131 scored clicks at a base conversion rate of 0.1361. Learning rates are selected per arm and per cell from a 7-point grid on a held-out slice of the initialisation period, before any streaming evaluation, and pinned across seeds.

Why one attribution window. We report $v = 1$ day only. A short click-attribution window is the common setting for click-through conversion optimisation, and a single window keeps every comparison origin-matched — arms are compared only against arms trained and scored on the same clicks against the same target. The cost is that our conclusions hold for short windows (Sect. 7).

Axes. Served capacity takes five values from 1,890 to 9,580,477 parameters, obtained by scaling the embedding tables and the interaction trunk together; the ladder spans more than three orders of magnitude so that a result holding at one model size alone cannot masquerade as general. *Write delay* o takes 5, 30 and 90 minutes, covering $F(o)$ from 0.355 to 0.803. Every arm is run at every combination with five seeds. The full one-day reproduction is 37 arms, 472 cells and 2,357 runs; this paper analyses the four axis arms of Sect. 3, the two references, and the frozen-teacher variants of Sect. 5.

Metric. We report normalised entropy,

$$\mathrm{NE} \;=\; \frac{\frac{1}{N}\sum_i \left[\, y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i) \,\right]}{\bar{y} \log \bar{y} + (1 - \bar{y}) \log(1 - \bar{y})}, \tag{3}$$

with $y_i = Y_i^{(v)}$, $\bar{y}$ the base rate, lower better. NE is pooled over all 8,196,131 scored clicks rather than averaged over rounds, since a per-round average normalises each hour by its own base rate and discards diurnal variation a model can legitimately predict. Our headline quantity is the difference ΔNE between two arms in one cell: absolute NE depends on corpus and capacity, whereas the question is which target wins at fixed everything-else.

Table 3. ΔNE at $v = 1$ d; negative favours the first arm. **Bold** marks significance at Holm-corrected $\alpha = 0.05$ over the 15 cells of each contrast. Five seeds per cell, paired.

Contrast	o	A0	A1	A2	A3	A4
Correct-Distill − Wait	5 min	**-0.0070**	**-0.0074**	-0.0024	-0.0005	-0.0015
	30 min	**-0.0087**	**-0.0090**	**-0.0043**	-0.0014	**-0.0057**
	90 min	**-0.0096**	**-0.0083**	**-0.0044**	**-0.0012**	-0.0003
Distill-Only − Wait	5 min	**-0.0018**	**-0.0030**	**+0.0023**	**+0.0057**	**+0.0056**
	30 min	**-0.0027**	**-0.0039**	**+0.0020**	**+0.0094**	**+0.0052**
	90 min	**-0.0039**	**-0.0041**	**+0.0015**	**+0.0091**	**+0.0051**
Correct-Distill − Distill-Only	5 min	**-0.0052**	**-0.0044**	**-0.0047**	**-0.0063**	**-0.0071**
	30 min	**-0.0060**	**-0.0050**	**-0.0064**	**-0.0108**	**-0.0108**
	90 min	**-0.0057**	**-0.0042**	**-0.0059**	**-0.0103**	**-0.0054**

Statistics. The five seeds differ only in initialisation and minibatch order — the data is identical across them — so seed i of two arms is a matched pair and we use a paired t-test within each cell, correcting across the 15 cells of a contrast with the Holm–Bonferroni step-down procedure. We never pool across cells: different o means a different label distribution. Seed dispersion is small at every rung (median NE standard deviation 0.00106, maximum 0.00241 over the arms studied), so cells are individually informative. Five seeds measure initialisation and shuffle variance on one data origin — precision, not generalisation — and we claim nothing beyond this corpus and split.

5 Results

We report ΔNE against Wait — negative means better than the admissible status quo — in Table 3 cell by cell, and in Fig. 1 against capacity.

The censored label is expensive; waiting is not a straw man. Wait beats Fresh by -0.2066 NE on average at $o = 5$ minutes, falling to -0.0118 at $o = 90$ — an ordering that follows $F(o)$, since the censored label is wrong for a $1 - F(o)$ share of eventual positives. This fixes the scale: the differences discussed next are one to two orders of magnitude smaller than the cost of getting the label wrong outright.

The hybrid target improves on waiting. Correct-Distill has lower NE than Wait in all 15 cells, 10 of them significantly, with gains from -0.0096 to -0.0003. The gain is largest where the served model is smallest and shrinks as capacity grows: averaged over write delays it is -0.0084 at A0 and -0.0025 at A4.

That decay is expected, but it bounds the claim. The teacher is fixed at 9,456,897 parameters — three orders of magnitude larger than the student at A0, comparable at A4 — so part of the small-capacity gain is the teacher's capacity delivered at no serving cost, which is what distillation is for, rather

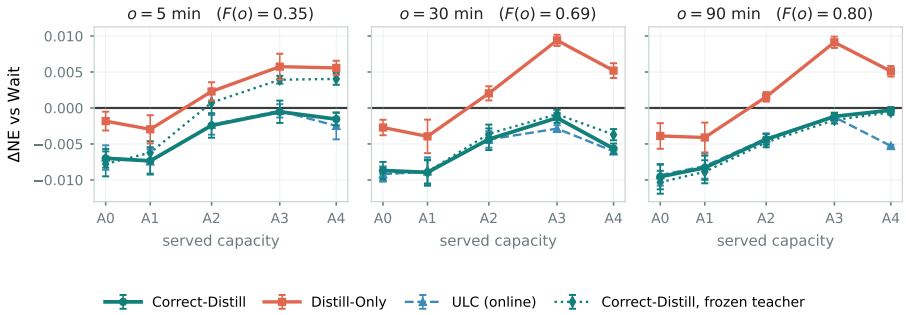


Fig. 1. ΔNE against Wait across the capacity ladder, one panel per write delay; below the line is better than waiting. Correct-Distill is below it everywhere and Ulc tracks it closely. Distill-Only crosses the line between A1 and A2 at every o. The frozen teacher matches the live one at $o = 30$ and 90 and separates from it at $o = 5$ for the larger students. Bars are the standard deviation of the five per-seed differences.

than anything specific to delayed feedback. What survives at A4, where student and teacher are the same size and the gain is still negative in all three cells, is the delayed-feedback component.

Removing the hard label does not. Distill-Only, our most informative result, beats Wait at the two smallest capacities (-0.0032 on average) and *loses* to it at the three largest (0.0051), crossing over between A1 and A2 at every write delay (Fig. 1). All 15 cells are significant, so the crossing is not noise.

The reading we find most plausible is that a purely soft target acts as a regulariser, whose benefit is a variance reduction that a low-capacity model can use and a higher-capacity one no longer needs [10]: at 1,890 parameters the model cannot exploit the hard label's extra information and gains from a smoother target, while by A2 it can, and discarding the realised outcome throws away signal it could now use. Against Correct-Distill the comparison is unambiguous: the hybrid wins all 15 cells by -0.0065 on average, so keeping the observed positive and softening only the negatives is strictly the better construction.

This cuts against the motivation of Sect. 6 rather than for it: a record carrying no hard label is the most convenient object for a multi-objective pipeline, and it is exactly the one that costs accuracy once the model is large enough to matter.

Parity with the online corrector. Correct-Distill and Ulc share Eq. (2) and differ only in how w is estimated. They are near-indistinguishable: mean ΔNE +0.00050, with 13 of 15 cells showing no significant difference. In the 2 cells that do differ, Ulc is the better of the two, by at most +0.00497. Plainly: *the offline teacher does not beat the online corrector*. Any case for the teacher route rests on what it does to the pipeline (Sect. 6), not on the metric.

How much of the teacher must stay live. Our default teacher keeps learning from the matured stream: a service, not an artifact. Re-running CORRECT-DISTILL with it frozen after causal initialisation, so that it never sees a matured record during the streaming period, costs +0.00092 NE on average (8 of 15 cells significant). The average hides the structure that matters. At $o = 90$ minutes freezing is free — the frozen teacher is in fact marginally *better*, -0.00050 — and at $o = 30$ it is nearly so. The cost concentrates in the hardest corner: at $o = 5$ minutes and the two largest capacities it reaches +0.00502. Frozen CORRECT-DISTILL still beats WAIT in 12 of 15 cells, failing only at $o = 5$ with a large student.

The teacher can therefore be a build-time artifact rather than a running service for most of the grid, which is the cheaper operating point; it has to stay live exactly where the pipeline is freshest and the model largest, because that is where its own initialisation goes stale fastest relative to what the student could otherwise learn.

Distance to the ceiling. For scale, ORACLE — which labels the fresh record with an outcome that has not happened — is better than CORRECT-DISTILL in 14 of 15 cells, by 0.0015 on average. The hybrid target therefore recovers much but not all of what perfect foresight would be worth, and the residual grows with capacity.

6 Toward Cross-Funnel Consolidation

The results above concern one objective; our motivation is several. This section states the connection and its limits; it is an argument, not a measurement.

Why heterogeneous windows block consolidation. Consolidating many ranking objectives into one model is an active industrial direction [11]. Consider one model serving K conversion objectives — a purchase, an app install, a lead — with attribution windows $v_1, \ldots, v_K$ differing by orders of magnitude. Under WAIT, objective k's record can only be written at $C_i + v_k$, when its label exists, so the K heads want records at K different times. A shared pipeline cannot honour that with one write per click: it must either emit a record per objective, which multiplies the stream by K and violates Rule 2 K times over, or delay every objective to $\max_k v_k$, which imposes the slowest funnel's staleness on the fastest.

What blocks consolidation is therefore the hard matured label, not the architecture.

What a write-time target changes. Both targets we advocate are computable at a single common write time. At $C_i + o$ the pipeline can evaluate, for every objective k simultaneously,

$$\tilde{y}_{i,k} \;=\; Y_{i,k}^{(o)} + \left(1 - Y_{i,k}^{(o)}\right) w_k^T(x_i), \tag{4}$$

because w_k^T is a function the pipeline can call rather than an outcome it must await. One record, written once at one time, can then carry all K targets. The heterogeneity in v_k is absorbed into the K teachers, each of which reconciles its own window offline, and the served pipeline becomes window-agnostic: it sees features and numbers.

What distinguishes the teacher from the online corrector. ULC also evaluates its corrector at write time, so it achieves the same alignment, and our results show it does so at the same accuracy. The difference is where the per-objective machinery lives. ULC fits each w_k as an auxiliary *inside* the served model's training loop, so consolidating K objectives means K auxiliaries coupled to one optimiser, each with its own maturity-gated input and its own failure modes. The teacher route places each w_k in a separate model upstream, whose only interface to the served pipeline is a number attached to a record. That is an argument about modularity, not about NE.

How far that reaches is decided by the frozen-teacher measurement of Sect. 5, and the answer is conditional. For write delays of 30 minutes or more the teacher can be frozen at negligible cost, so each w_k becomes a build-time artifact — a model trained once, shipped, and queried — and consolidating K objectives adds K artifacts rather than K live maturity-gated services. At a five-minute pipeline with a large served model the teacher must stay live, and the teacher route's operational advantage over ULC narrows to decoupling alone.

What this evidence does not establish. We ran one objective, so we have no evidence about interference between objectives sharing a trunk — the central risk of consolidation, and what a multi-task study would have to measure. We have no evidence that K teachers are affordable, only that one is: teacher capacity is training-only but not free, and 9,456,897 parameters per objective is a real cost. And our single objective has a short window, the easy case; at $v = 28$ days the teacher's own labels are far staler, and the argument would need re-testing rather than extrapolating.

What we claim is narrower and, we think, still worth reporting: under a constraint that eliminates most published methods, a target computable at write time is available that costs nothing against the status quo, and the most pipeline-convenient version of it — the one that carries no hard label at all — is the one that does not work. That is a negative result cheaper to learn on one objective than on K.

7 Conclusion

Imposing one production constraint — at most one training record per click, its label frozen when written — eliminates 8 of 11 published delayed-feedback methods, several by wide margins, and leaves a design space whose only remaining choice is the target the single record carries. Along that axis at a one-day window: waiting for the matured label far beats training on the censored one; an

offline teacher's estimate substituted for part of the hard label improves on waiting in every cell; removing the hard label entirely does not, crossing from better to worse as served capacity grows; and the hybrid matches an online corrector rather than beating it, so the case for a teacher is architectural.

Limitations. One dataset and one temporal split. One attribution window, deliberately the short-window case where the teacher's job is easiest. And a teacher fixed at a capacity comparable to our largest student, so small-capacity gains partly reflect distillation from a larger model rather than delayed-feedback correction; the effect surviving at matched capacity is the smaller one.

Next. The experiment this is a precondition for is the multi-objective one: K heads with K windows on one trunk, one record per click, per-objective teachers upstream. It must answer whether objectives interfere when sharing a trunk, and whether K teachers are affordable. Our result says the target construction is sound enough to make the question worth asking.

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